

Tide retrospective: the fourth-cumulant limit (gamma-rung programme, stages 4–5)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-09

1 Setting

The programme’s summit. Stages 1–3 built, in order: the signed exponential remainder, the second-order rescaled integral expansions, and the Gibbs moment rates. This tide assembles them into the limit the whole programme exists to prove.

2 The thing formalised

Three declarations in `Laplace/OneD/Kappa4Asymptotic.lean`, sorry-free with zero warnings: two private squeeze converters (an order-two rate $|f - L - C'/t| \leq K/(t\sqrt{t})$ gives $t(f - L) \rightarrow C$; a $\sqrt{t}$ -rate gives a plain limit) and the theorem¹: for the anharmonic potential with the standing hypotheses, writing μ_r for the Gibbs expectation of x^r ,

$$t^3(\mu_4 - 4\mu_3\mu_1 - 3\mu_2^2 + 12\mu_2\mu_1^2 - 6\mu_1^4) \longrightarrow \frac{3\alpha^2}{\lambda^5} - \frac{\gamma}{\lambda^4} \quad (t \rightarrow \infty).$$

The moment combination is the fourth cumulant; the limit was pinned numerically at programme scoping and its cancellation table verified symbolically before stage 1.

3 Proof strategy

Pure limit algebra over the stage-3 rates. The split

$$t^3\kappa_4 = t\left(t^2\mu_4 - \frac{3}{\lambda^2}\right) - 3t\left((t\mu_2)^2 - \frac{1}{\lambda^2}\right) - 4(t^2\mu_3)(t\mu_1) + 12(t\mu_2)(t\mu_1)^2 - 6\frac{(t\mu_1)^4}{t}$$

is one **ring**; the first piece converges to C_4 by the squeeze, the second factors as a difference of squares and converges to $2C_2/\lambda$, the third and fourth are products of first-order limits (the seabed’s mean asymptotic supplying $t\mu_1$), and the last vanishes. The t^{-2} Gaussian contributions cancel structurally between μ_4 and $3\mu_2^2$; the constant evaluation $(C_4 - 6C_2/\lambda - 180A^2/\lambda^2 + 108A^2/\lambda^2 = 3\alpha^2/\lambda^5 - \gamma/\lambda^4)$ is the radical-atom substitution and **ring**.

¹`kappa4.anharmonic.asymptotic`.

4 Roadblocks and resolutions

Six build iterations, every one a documented catalogue class: `squeeze_zero_norm` takes positional arguments; `Real.sqrt_atTop` is spelled `Real.tendsto_sqrt_atTop`; two dead rings after `field_simp`; one `field_simp` needing its explicit nonzero hypothesis; and `tendsto_const_nhds` needing the type-ascribed anchor — the catalogue’s own entry, which cost one iteration instead of a debugging session.

5 What was Mathlib, what was new

Mathlib: `Tendsto` algebra and the squeeze. Seabed: everything — four stages of this programme and the original Stage-2 mean asymptotic, composed and consumed. New: only the assembly.

6 Lessons

A programme scoped with a numerically pinned target and a symbolically verified route runs its summit tide as pure bookkeeping: nothing mathematical moved between the scoping consult and this proof. The squeeze converters are the reusable shape for turning any of the seabed’s rate bounds into limits.

7 Follow-ups

- Stage 6 (final): the gamma recovery corollary — eventual agreement of the observable data pins γ given (λ, α) , completing the anharmonic jet of the germbij note’s Theorem 3.1 and earning its marker.